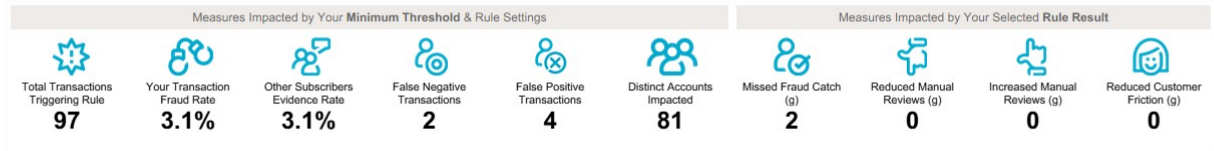




Geolocation and IP Mismatch

Rule Tuning Scenario Impact



Filters



Navigation



Legends

- Has IP Mismatch
- No Mismatch
- Has Mismatch

Rule Tuning Scenario Tests

IP Address Mismatch

The **IP Address Mismatch Rule** triggers when the subscriber's IP address (Stated IP) does not match the Real IP Address (provided by TransUnion). Usually when there is a mismatch, the consumer is using a proxy that masks or changes the IP address.

In some cases, the IP addresses are different but the geolocation reflects the same location (e.g. city, state, country). Since the rule triggers simply by checking whether or not there is difference between the two IPs, be sure to examine the false positive rate at your selected integration point to ensure the transactions triggering the rule are targeting **high-risk behavior** with minimal impact to **good customers**.

Geolocation Mismatch

Use the **Geolocation Mismatch (GeoMM) Rule** to determine if the location mismatch has significance. The GeoMM rule allows you to change the Mismatch Level, to invert the rule, and to specify a set of countries to exclude from the risk check. When determining if this anomaly should be targeted with a rule, you will **select the geolocation level** (e.g. Country, Region/State, or City) as the precision.

There must be an IP mismatch present in order to use the Geolocation Mismatch rule.

Select Mismatch Level

- ☒ Country
- ☐ Region / State
- ☐ City
- ☐ Any (IP Mismatch)

Desired Rule Result

- ☒ Deny
- ☐ Review
- ☐ Allow

IP Address Mismatches (No Level)

7,511
Transactions with an IP
Address Mismatch

Distinct Accounts

Distinct Devices

160,777

7,511

No Mismatch

Has Mismatch

Geolocation Mismatches by Level

	Total Transactions Triggering Rule	Your Transaction Fraud Rate	Distinct Accounts Impacted	Distinct Devices	False Positive Transactions	False Negative Transactions
Country Level	97	3.1%	81	79	4	2
Region Level	3,720	3.2%	2,987	2,977	196	72
City Level	5,671	4.2%	4,523	4,507	303	164

Integration Point Impact & Detail

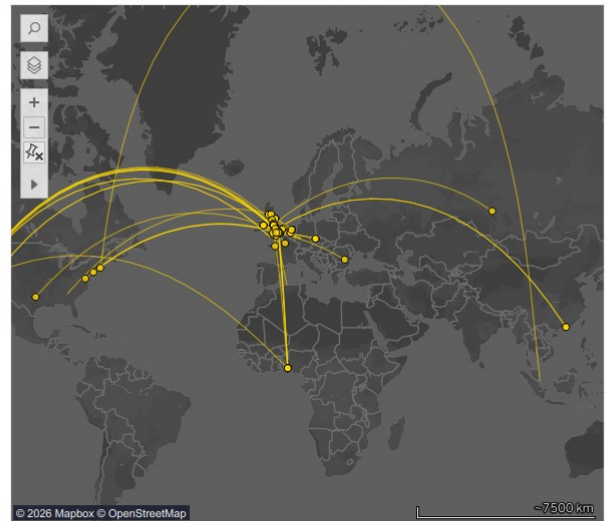
Summary of Rules Triggered by Integration Point

Hover to view more Impact metrics.

INTEGRATION_POINT_NAME	Total Transactions Triggering R...	Your Transaction Fraud Rate	Other Subscribers Evidence Rate	False Positive Transactions	False Negative Transactions
enrollment	97	3.1%	3.1%	4	2

Map of Transactions with Geolocation Mismatch for Selected Level

Hover on distance lines to view more information. Click to view transaction details.





Blackbox Age Analysis

Rule Tuning Scenario Impact

Measures Impacted by Your Minimum Threshold & Rule Settings						Measures Impacted by Your Selected Rule Result			
Total Transactions Triggering Rule	Your Transaction Fraud Rate	Other Subscribers Evidence Rate	False Negative Transactions	False Positive Transactions	Distinct Accounts Impacted	Missed Fraud Captured (bb)	Reduced Manual Reviews (bb)	Increased Manual Reviews (bb)	Reduced Customer Friction (bb)
12,013	4.6%	5.6%	406	1,277	9,401	406	18	0	0

Rule Tuning Scenario Tests

The **Blackbox Age Rule** will trigger for any transaction where time that has passed between the creation of a blackbox (e.g. page load) and the submission of a transaction is more or less than the threshold you specify below.

Because a blackbox is typically created at or near the time that the transaction is submitted, this rule targets risky behavior when the elapsed time is **unusually long** (suspicious attempts to reuse blackboxes) or **unusually short** (potential botnet or scripting activity). **We do not recommend setting the weight high enough that this rule on its own will deny transactions.**

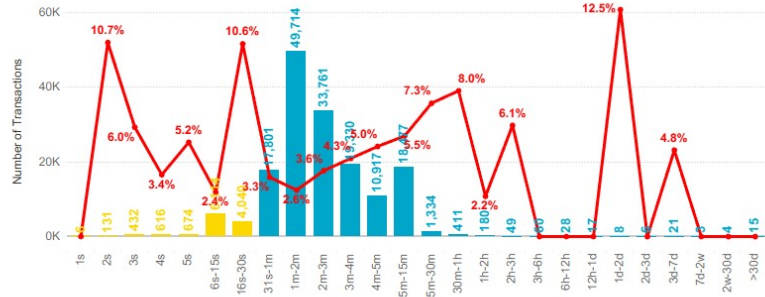
Set Rule Minimum Threshold. Use drop-downs to change time dimension and direction. (Bins in Yellow Trigger Rule)

<= 30 seconds = 30 seconds

Desired Result
☒ Deny
☐ Review
☐ Allow

Blackbox Age Distribution with Other Subscriber Evidence Rate Trends

Selected Blackbox Age: <= 30 seconds.



Integration Point Impact & Detail

Summary of Rules Triggered by Integration Point

Hover to view more Impact metrics.

INTEGRATION_POINT_NAME	Total Transactions Triggering R..	Your Transaction Fraud Rate	% Devices with OtherSub Evidence	False Positive Transactions	False Negative Transactions
enrollment	12,013	4.6%	6.0%	1,277	406

Selected Transactions from Primary Chart for More Info

Click histogram bin to view more details.

Transactions with Negative or Undetermined Blackbox Ages

Click row to view the transaction detail.

TRACKING_NUMBER	TXN_DATE	New Device (to Sub)?	Blackbox Age (seconds)	
220406143007545197	6/29/2021	No	Null	✓
639192524227970598	6/29/2021	No	Null	✓
430792040423635067	6/29/2021	No	Null	✓
158188255005225602	6/29/2021	No	Null	✓
162504300266682001	6/29/2021	No	Null	✓
226230766806021536	6/29/2021	No	Null	✓
150823554154336693	6/29/2021	No	Null	✓
250431561688274875	6/29/2021	No	Null	✓
112818066767424595	6/29/2021	No	Null	✓
466840309106570622	6/29/2021	No	Null	✓
156848387943702861	6/29/2021	No	Null	✓
171546662588410869	6/29/2021	No	Null	✓
156432137222578796	6/29/2021	No	Null	✓
711829550742069830	6/29/2021	No	Null	✓
681779010549771351	6/29/2021	No	Null	✓
559299064554903420	6/29/2021	No	Null	✓
262011951202124767	6/29/2021	No	Null	✓
212596186851310424	6/29/2021	No	Null	✓
648251603903723476	6/29/2021	No	Null	✓
276633062476951921	6/29/2021	No	Null	✓
147153870784043356	6/29/2021	No	Null	✓
247179972475342992	6/29/2021	No	Null	✓
496575482769340644	6/29/2021	No	Null	✓
205317701804295707	6/29/2021	No	Null	✓
163569631989222796	6/29/2021	No	Null	✓
197424437913050375	6/29/2021	No	Null	✓
108991133194485910	6/29/2021	No	Null	✓
253117604410240306	6/28/2021	No	Null	✓
303043380760923193	6/28/2021	No	Null	✓
701582226380702354	6/28/2021	No	Null	✓
561417502232423129	6/28/2021	No	Null	✓
278876650636067686	6/28/2021	No	Null	✓
334644039389322552	6/28/2021	No	Null	✓
871882920199850353	6/28/2021	No	Null	✓
286323957212229957	6/28/2021	No	Null	✓
679092110686358533	6/28/2021	No	Null	✓
944485149830059422	6/28/2021	No	Null	✓
241598227777052721	6/28/2021	No	Null	✓

Filters



Navigation



Legends

Transaction Result
✓ allow
△ review
♦ deny



SureScore (Machine Learning Device Model)

Rule Tuning Scenario Impact

Measures Impacted by Your Minimum Threshold & Rule Settings					Measures Impacted by Your Selected Rule Result				
Total Transactions Triggering Rule	Your Transaction Fraud Rate	Other Subscribers Evidence Rate	False Negative Transactions	False Positive Transactions	Distinct Accounts Impacted	Missed Fraud Catch (ss)	Reduced Manual Reviews (ss)	Increased Manual Reviews (ss)	Reduced Customer Friction (ss)
286	2.1%	0.7%	1	211	281	1	1	0	0

Rule Tuning Scenario Tests

The **SureScore Range Rule** will trigger for any transaction that fall within a specific range of ML Device Model values. You can define maximum and minimum thresholds for transactions, and add to the transaction total when the score is between these. **Scroll down for more uplift opportunities when using this rule.**

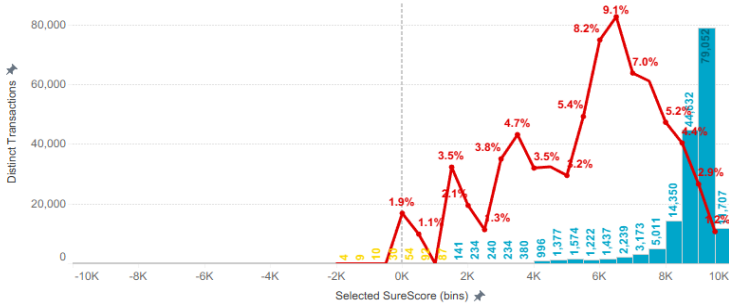
Minimum Threshold: Maximum Threshold: Desired Rule Result: ☒ Deny ☐ Review ☐ Allow

Chart Configuration

Select Score Type: ☒ Global ☐ Vertical ☐ Either Change the size of the bins:

SureScore Distribution with Other Subscriber Evidence Rate Trend

Selected Score Range: Minimum >= -10,000 | Maximum <= 1,500.



Integration Point Impact & Detail

Summary of Rules Triggered by Integration Point

Hover to view more impact metrics.

INTEGRATION_POINT_NAME	Total Transactions Triggering R...	Your Transaction Fraud Rate	Other Subscribers Evidence Rate	False Positive Transactions	False Negative Transactions
enrollment	286	2.1%	0.7%	211	1

Selected Transactions from Primary Chart for More Info

Click transaction row below to view even more details.

Additional Rule Considerations

These scenarios allow you to target and impact customers with extreme scores.

Understanding the Data

SureScore uses advanced predictive machine learning to uncover trusted customers, improve the online experience, reduce costly step-ups, and prevent more fraud by exposing hidden risks. While the verified phone number and email address can be extremely valuable when assessing the customer's **identity risk**, SureScore provides a score on both new and returning customers based on **all aspects** of the transaction. Thus, our trust service helps you:

Decrease manual review queues as much as 40 percent by ensuring low risk transactions from legitimate customers aren't stuck in your review queue.

Create good user experiences by allowing your best customers to skip costly security checks while building loyalty with a frictionless customer experience.

Increase revenue opportunities by using trust scores at account creation to offer special promotions, or immediately convert a customer to a high status VIP.

See the **Avoidable Manual Review Costs** metric showcasing those legitimate customers for whom you can create a positive experience by avoiding the delay of manual review.

Avoidable Manual Review Costs

246

Transactions

with a high SureScore range where the customer was delayed by manual review. Our proprietary score ensures that low risk transactions from legitimate customers aren't creating customer friction and increasing operational costs.

Test VIPs Stuck in Review



Missed Fraud Catch

Transactions

with very negative risk scores that were allowed to proceed. Business rules allow you to target known suspicious patterns and enforce regulatory restrictions. SureScore supports your fraud catch by examining undiscovered patterns.

Test DENY Range (Negatives)



Filters



Navigation



Legends